Final Project Report

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1. Introduction

Electric Vehicle Market Performance & Insights uses Tableau to present selected indicators from an electric-vehicle dataset. The purpose is to make EV market and battery measures easier to explore in one workbook.

1.1 Project purpose

The dashboard supports descriptive exploration of manufacturer sales, annual sales trends, battery technology, customer ratings, battery capacity, degradation, and cycle life. It does not establish causation or provide validated forecasts.

2. Problem statement

EV market information can be difficult to compare when measures are spread across separate tables or views. This project organizes selected measures into eight Tableau charts across two dashboards.

3. Requirements

Load the supplied dataset, create eight worksheets, arrange them into two dashboards, add navigation, publish the workbook, and preserve the packaged workbook. Describe additional controls only if verified.

4. Dataset and technology

The project uses **ev_market_test.csv** and Tableau Public/Desktop. Fields used include Manufacturer, Model Year, Battery Type, Vehicle Id, Annual Sales Units, Customer Rating, Battery Capacity Kwh, Battery Degradation Pct 5Yr, and Battery Cycle Life. Confirm field definitions and aggregations against the dataset and worksheet configuration.

5. Implementation — eight worksheets

• Top 10 EV Manufacturers — annual sales units by manufacturer, limited to the top ten.

• Battery Technology Distribution — vehicle-record count by battery type.

• Annual EV Sales Trend — annual sales units by model year.

• Average Customer Rating — average rating.

• Average Battery Capacity — average capacity by battery type.

• Battery Degradation Analysis — five-year degradation measure across model years.

• Average Battery Cycle Life — average cycle life by battery type.

• Battery Degradation by Type — average five-year degradation by battery type.

6. Dashboard organization

Dashboard 1 groups worksheets 1–4. Dashboard 2 groups worksheets 5–8. Navigation buttons allow viewers to move between dashboards.

7. Testing and validation

The workbook was published to Tableau Public and a packaged .twbx workbook was saved. Final checks should include opening both dashboards, testing both navigation buttons, checking chart labels and values against source data, and confirming the published link is accessible. No measured load-time or user-study results are claimed.

8. Results, advantages, and limitations

The implemented output is an eight-chart, two-dashboard Tableau workbook. Advantages include a consolidated view and separation of market and battery analyses. Limitations include dependence on dataset coverage, freshness, completeness, consistent units, and field definitions.

9. Conclusion

The project demonstrates how Tableau can organize selected EV market and battery indicators into a dashboard experience. Conclusions should be drawn only after validating displayed figures and understanding dataset limitations.

10. Future scope

Possible extensions include verified charging-infrastructure and regional data, refresh automation, validated forecasting, and usability testing.

11. Appendix and evidence

Dataset: ev_market_test.csv
Workbook: packaged Tableau .twbx saved locally
Tableau Public URL: add verified link
GitHub URL: add after repository creation
Demo recording: add after recording
Screenshots: add actual screenshots of both dashboards and key worksheets.